



# Investigation report

## Rejected take-off and runway overrun, Tarlton Springs - ATSB-style final report

Drafted 07/10/2026

INVESTIGATION WORKFLOW SUITE

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## Investigation summary

### What happened

On 18 June 2026, Saltbush Air flight SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off from runway 09 at Tarlton Springs Aerodrome at 17:30 local time. This was 5 minutes after last light, with no moon. On board were 44 passengers, 2 flight crew and 1 cabin crew. At 116 kt, 2 seconds after V1, the first officer saw two kangaroos crossing onto the runway from the north. The captain rejected the take-off at 118 kt, 6 kt above V1, with 695 m of sealed runway remaining. The aircraft struck both animals 1 second later. Reverse thrust was then available on the No. 1 engine only.

The aircraft left the runway end at approximately 21 kt. It stopped upright on the extended centreline, 62 m into the 90 m runway end safety area. There were no serious injuries. One cabin crew member reported bruising during the disembarkation, and both kangaroos were killed. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond repairable limits, the right main gear doors were distorted, both right main tyres deflated and the lower right fuselage was punctured. The aerodrome was closed for 4 hours 20 minutes.

### What the investigation found

The operator required a runway inspection before the first departure after last light only where a company aircraft had reported a wildlife strike in the preceding 24 months. Tarlton Springs had never met that criterion, although its log recorded eleven kangaroo sightings on the sealed runway. No inspection was requested before SB512 departed. Nothing in the operator's documents prompted the crew to consider one.

The evening service was introduced as a schedule change only. The route risk assessment, which restricted Tarlton Springs to daylight operations, was not reopened. SB512 therefore departed in darkness during the logged peak of kangaroo movement, and the crew could not see the animals until they reached the lit runway.

The kangaroos entered through a failed section of the northern perimeter fence. The aerodrome operator had deferred its repair at three budget rounds. The aerodrome fell below the threshold at which the aerodrome standard requires a documented wildlife hazard management plan, so no hazard assessment stood behind those deferrals. That link was less firmly established than the other contributing factors.

Rejecting above V1 shortened the runway left for the stop. The operator's training and procedures had no scenario or guidance for a runway obstruction during the take-off roll. The captain stated that he knew the procedure and departed from it deliberately. The strike then reduced deceleration to about 15% below the certified value. Both effects were needed to take the aircraft past the runway end. Maximum braking was established 2 seconds after the reject decision, against a fleet mean of 3.2 seconds, which reduced the distance travelled into the safety area.

Two further safety issues did not contribute. The evening rotation was rostered to close five-sector duties within 45 minutes of the duty limit. Floor escape path lighting on the type was checked only every 24 months and not before flight. The investigation could not determine whether the crew were fatigued, or whether four runway edge lights brighter than nominal affected detection of the animals.

### What has been done as a result

On 25 June 2026 the aerodrome operator installed a temporary electric wire along the failed fence section. It funded the fence replacement and programmed it for September 2026. On 26 June 2026 the operator listed Tarlton Springs in Appendix 8A by temporary revision TR-2026-04, which requires a runway inspection before every departure there after last light. The investigation recommended that the operator list aerodromes on wildlife exposure rather than a reported strike. It also recommended a route risk assessment trigger for any aerodrome first operated into after last light. The operator's response was due by 30 September 2026.

## Safety message

A control armed only by a reported strike cannot protect against the first one. Wildlife precautions should rest on the exposure that aerodrome logs already show. Moving an operation into darkness should prompt a check that its risk assessment still holds. Crews departing at night can request a runway inspection where an aerodrome offers one.

## The occurrence

All times in this section are UTC. Local time at Tarlton Springs was UTC plus 10 hours.

### Before departure

On 18 June 2026, Saltbush Air flight SB512 was to operate from Tarlton Springs Aerodrome to Port Carrick. The aircraft was a 50-seat twin-turboprop registered VH-SBQ, and this was the last scheduled sector of the day. Tarlton Springs has no control tower, so crews make their own broadcasts on the common traffic advisory frequency (CTAF).

The aerodrome operator carried out its daily movement area inspection at 21:35 on 17 June, which was 07:35 local on 18 June. This was before the first scheduled arrival, and the inspection recorded the movement area as clear. It was the only runway inspection that day. No further inspection was carried out or requested.

The flight crew had signed on at 06:30 local for a rostered three-sector duty. That morning, a member of the crew rostered for the evening Tarlton Springs rotation reported unfit. The rotation was added to this crew's duty, which made SB512 their fifth sector. During the third sector a ground delay at Port Carrick was recovered on the Tarlton Springs turnaround. The doors closed at 07:16, and the aircraft departed on schedule against a scheduled off-blocks time of 17:20 local. On board were 44 passengers, 2 flight crew and 1 cabin crew.

The first officer was the pilot flying. She had flown into Tarlton Springs twice before, both times in daylight. Both pilots told the investigation that animals on the aerodrome were not discussed at any point during the turnaround or the brief. The calculated V1 was 112 kt. It was based on a weight of 20,800 kg, flap 10, a dry runway 09 and a 5 kt headwind component.

At 07:21:50 the crew activated the pilot-activated runway lighting with three transmitter keyings. The captain then remarked, "Lights are on. That's thirty minutes on the timer." The crew broadcast their taxi call at 07:23:15. At 07:28:51 the first officer reported the AWIS wind as "zero nine zero at six". At 07:29:10 the captain broadcast that SB512 was lining up on runway 09 for departure to Port Carrick. The crew did not request a runway inspection, and the aerodrome did not offer one. No wildlife transmission had been made on the CTAF in the preceding 90 minutes.

### The take-off roll and rejected take-off

Power was set at 07:30:02, with the engines at 98.4% and 98.5% torque. The aerodrome was in darkness, as the roll began 5 minutes after last light. The 80 kt cross-check was made at 07:30:15 with no caution or warning active. Throughout the roll the recorded headwind component was 4 to 6 kt. The captain called V1 at 07:30:29, at 112 kt indicated airspeed.

At 07:30:31, at 116 kt, the first officer called kangaroos ahead: "roos, roos, on the left". According to the occurrence notification, two kangaroos were moving onto the runway from the northern side, past the 1,100 m point. In interview the first officer estimated their range at first sight as about 200 m, and said it could have been less. After the aircraft stopped, she told the captain on the cockpit voice recorder that the animals "came out of the dark on the left of the strip, crossed the centreline". The captain replied that he "never had them" because he was "heads-in on the engine page after V one".

At 07:30:32, at 118 kt, the captain called "Stopping. I have control." Both power levers reached flight idle within 0.6 seconds. The reject was therefore initiated 3 seconds and 6 kt after V1. At that point the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining.

About one second later, at 07:30:33 and 114 kt, the aircraft struck both animals. The flight data recorder showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The impact was taken by the right main landing gear and the No. 2 propeller. The carcasses were later recovered at 1,188 m and 1,221 m from the runway 09 threshold, and animal tissue was found in two No. 2 propeller blade roots. From the moment of impact, No. 2 propeller RPM decayed.

Maximum braking was established at 07:30:34. This was 2 seconds after the decision, against an operator fleet mean of 3.2 seconds. Brake pressure was 2,940 psi, with anti-skid active on all four wheels. At 07:30:36 reverse was selected on No. 1 engine only. The No. 2 reverse interlock did not make, and the captain called that "number two's not coming". Between 07:30:34 and 07:30:44 the mean deceleration was 2.6 m/s<sup>2</sup>. That was approximately 15% below the value used in the certified accelerate-stop data, which assumes symmetric reverse. At 07:30:47 the captain said, "We're not stopping on it." Continuous tyre marks ran from 1,160 m to the runway end.

## The overrun

The aircraft left the end of runway 09 at 07:30:51, at approximately 21 kt indicated airspeed (20 kt groundspeed). In the graded gravel, longitudinal deceleration fell to 1.9 m/s<sup>2</sup>. The aircraft stopped at 07:30:57, 62 m into the 90 m runway end safety area. It came to rest upright and on the extended centreline, with the nose wheel in graded gravel. This was 28 m short of the end of the prepared safety area, beyond which the terrain is ungraded. From 118 kt the aircraft had needed 757 m to stop, against the 695 m of sealed surface that had remained.

## After the aircraft stopped

At 07:31:04 the captain set the parking brake. He noted no fire indications and nothing on the warning panel, and he stated that he was not evacuating. The first officer agreed. Both fuel levers were moved to cut-off at 07:31:26. At 07:31:34 the captain told the cabin to remain seated.

At 07:31:48 the aerodrome reporting officer, who had seen the aircraft go off the end of the runway, called on the CTAF. The captain reported 44 passengers and three crew, with no injuries and no fire. He requested activation of the aerodrome emergency plan and a vehicle. The reporting officer activated the plan at 07:32:10 and dispatched a vehicle. At 07:32:24 the reporting officer broadcast that the runway was obstructed and the aerodrome was closed.

The forward airstair was heard operating at 07:34:40. No evacuation was commanded and no slides were deployed. The main cabin lights remained on throughout. The occurrence notification states that passengers were disembarked 4 minutes 8 seconds after the aircraft stopped. The timeline, however, places the disembarkation from 07:34:40 to 07:38:48. That period is the same length, but it begins when the airstair was operated rather than when the aircraft stopped. During the disembarkation the first officer noticed unlit sections of the aisle floor strip towards the rear of the cabin. At the time, the emergency lighting was armed and the ceiling lights were on. She raised this with the engineer the next morning.

There were no serious injuries. One cabin crew member reported bruising during the disembarkation. Both kangaroos were killed.

The aircraft was substantially damaged. Three No. 2 propeller blades were damaged beyond repairable limits, and the engine required a shock-load inspection. The right main gear doors were distorted. Both right main tyres deflated through the fuse plugs, with brake energy calculated at 84% of the certified maximum. Propeller debris punctured the lower right fuselage.

The aerodrome was closed for 4 hours 20 minutes. Both flight crew were stood down and tested for drugs and alcohol, and both results were negative. The operator's engineering representative

removed the FDR and CVR at 12:40. The Head of Flying Operations suspended the evening Tarlton Springs service.

## Context

### Personnel information

#### Captain

The captain held 8,410 hours total, 2,140 on type, and had been in command for 3 years and 1 month. He had flown one other turboprop type and held no jet ratings. His last cyclic simulator check (11/03/2026) and line check (02/12/2025) were satisfactory, and he had no prior occurrence history. His duty and rest in the preceding 7 days were within company limits with no variations. He had two rest days before 18/06/2026.

#### First officer

The first officer held 1,655 hours total, 405 on type, and had been with the company for 14 months. Her last cyclic simulator check, on 28/04/2026, was satisfactory on all items. She had flown two previous daylight sectors into Tarlton Springs, on 09/01/2026 and 17/02/2026. She had one rest day before 18/06/2026.

#### Duty on the day

Both pilots signed on at 06:30 local for three sectors: Port Carrick to Hensby, Hensby to Mardle Creek, and Mardle Creek to Port Carrick. On the morning of 18/06/2026, a rostered crew member reported unfit, and the evening rotation (SB511 Port Carrick to Tarlton Springs and SB512 Tarlton Springs to Port Carrick) was added to their duty. The occurrence was 11 hours 00 minutes into a planned duty of 13 hours 00 minutes, finishing at 19:30 local. The approved fatigue risk management system permits 13 hours 30 minutes for five sectors from that sign-on time.

The rotation operated on 289 days between 14/07/2025 and 18/06/2026. It was always rostered as the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00 and planned at 12 hours 45 minutes to 13 hours 15 minutes. On six of those days, it was reassigned to a crew already on duty after a rostered crew member reported unfit, extending their day from three sectors to five. 18/06/2026 was the sixth such day.

In interview, both pilots described themselves as tired at take-off, in a way they considered ordinary for a five-sector day. The first officer said she had taken a proper break in Port Carrick mid-duty.

### Aircraft information

VH-SBQ was a 50-seat twin-turboprop operated by Saltbush Air. [NOT ESTABLISHED: aircraft manufacturer, model, serial number and year of manufacture.] At 22,900 kg, the type is the heaviest scheduled aeroplane at Tarlton Springs. Its two engines, No. 1 and No. 2, each have a reversing propeller, and the scheduled accelerate-stop figures assume symmetric reverse.

After the occurrence, a borescope inspection of both engines found no pre-existing damage and no evidence of a power anomaly before the impact. The anti-skid system tested serviceable. The right main gear structure was within limits on detailed inspection.

The pre-flight emergency lighting test illuminates the ceiling emergency lights and exit signs but not the floor proximity escape path segments. Those are checked only by a functional test every 24 months, last done on VH-SBQ 11 months before the occurrence.

The first officer reported unlit sections of the aisle floor strip. A functional check on 24/06/2026, with the system armed on battery power, found two of the six left-side floor proximity segments aft of row 16 did not illuminate. The cause was a failed ballast common to both. No check available

to the crew or a line engineer could have detected it. [NOT ESTABLISHED: whether the same ballast part is fitted across the operator's fleet of the type.]

## Meteorological information

The Tarlton Springs automatic weather station reported:

- METAR XTLS 180630Z AUTO 09007KT 9999 FEW040 09/M01 Q1024 RMK RF00.0/000.0  
[weather observations]
- METAR XTLS 180700Z AUTO 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0  
[weather observations]
- METAR XTLS 180800Z AUTO 09005KT 9999 FEW040 07/M02 Q1024 RMK RF00.0/000.0  
[weather observations]

These gave a headwind component of 5 to 7 kt on runway 09. The crew's forecast was "09008KT 9999 FEW040", becoming "08006KT CAVOK" from 10:00 UTC. The runway was dry. The performance calculation used a 5 kt headwind component.

Sunset was at 16:58 local and civil twilight ended at 17:25 local. The moon, a waning crescent 14% illuminated, rose at 22:41 local. No town lighting was visible from the aerodrome. On 23/06/2026, visiting at the same time of day, the investigator could not see the unlit graded strip either side of the sealed surface from the runway centreline.

## Recorded information

The aircraft carried a flight data recorder (FDR), a quick access recorder (QAR) and a cockpit voice recorder (CVR). The operator's engineering representative removed the FDR and CVR at 12:40 UTC on 18/06/2026. The operator's Flight Data Section issued report FDA-2026-0417-R1 on the FDR and QAR data on 24/06/2026. The CVR (CVR-2026-0417) and the Tarlton Springs CTAF recording were transcribed on 25/06/2026 and verified by a second investigator.

The FDR data covers the take-off roll, stop and engine shutdown, and the stopping distance was reconstructed from recorded groundspeed. The Flight Data Section compared the crew's response time with the fleet distribution for rejected take-offs, whose mean from decision to maximum braking was 3.2 seconds.

## Aerodrome information

Tarlton Springs is a certified aerodrome without a control tower, where traffic broadcasts are made on the CTAF. It handled 61,400 passenger movements in 2025. Runway 09/27 is sealed within a graded strip. The accelerate-stop distance available (ASDA) for runway 09 was 1,847 m. Beyond the runway end lies a 90 m runway end safety area (RESA) of graded gravel, then ungraded terrain.

## Runway lighting

The pilot-activated lighting is keyed from the CTAF and runs for 30 minutes at a single intensity set at the lighting controller, which crews cannot vary. The controller was at stage 3, the highest of three stages and the commissioning setting.

A photometric check on 25/06/2026, requested by the investigator, found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal stage 3 intensity. The rest were within tolerance. The works register and electrical log show no work on the lighting or controller between 18/06/2026 and the check. The maintenance programme checks lamp serviceability but not intensity.

## Wildlife hazard management

Aerodrome Standard AS-139 section 10.4 requires a certified aerodrome operator to "take reasonable measures to reduce the risk to aircraft operations presented by wildlife". Subsection 10.4.2 requires a documented wildlife hazard management plan only where aeroplanes above 27,000 kg maximum take-off weight use the aerodrome, or it serves more than 350,000 passenger movements in a calendar year. Otherwise, under 10.4.3, measures "need not be documented", but the operator "must be able to demonstrate what measures are in place".

Tarlton Springs was below both thresholds, held no plan, and was not in breach. Its measures were a daily inspection before the first arrival (Aerodrome Manual 6.3) and a wildlife sighting log. Aerodrome Manual 6.3 also allows unscheduled inspections at the request of the reporting officer or a pilot in command. The reporting officer was at the terminal during the SB512 turnaround and confirmed an inspection would have been done if requested. Aerodrome staff go off duty at 18:30 local.

Works register item WR-221 (04/2023) recorded displaced panels and strained wire in the northern perimeter fence between the 700 m and 1,300 m markers, with animals passing through. It recommended replacement at an estimated \$41,000, which was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263 (02/2026) recorded further deterioration. On 23/06/2026 the investigator found a continuous 140 m section with the bottom two wires separated from the posts, and a clear animal pad through the fence at the 1,050 m mark (attachments/Failed\_northern\_fence\_left\_unrepaired\_through\_three\_budget\_rounds\_1.jpg).

## Organisational and procedural information

### Wildlife procedures

Operations Manual Part A 8.14.1 states that wildlife is a hazard "particularly at regional aerodromes and particularly around dawn and dusk". Under 8.14.2, at an Appendix 8A aerodrome, the pilot in command "shall request a runway inspection from the aerodrome operator prior to the first departure after last light". Elsewhere, under 8.14.3, "no specific action is required".

Appendix 8A lists an aerodrome "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months". On 18/06/2026 it listed nine aerodromes, never including Tarlton Springs, and was last reviewed on 03/03/2025. Part A neither asks crews to report sightings nor provides any other route for adding an aerodrome. The first officer stated that the flight plan, charts, aerodrome directory entry and company aerodrome brief page for Tarlton Springs did not mention animals.

### Rejected take-off procedures and training

Operations Manual Part B 4.6.3 limits rejection between 80 kt and V1 to listed conditions, including the pilot in command's judgement that the aircraft is unsafe or unable to fly. Section 4.6.4 states "At or above V1 the take-off shall be continued." Section 4.6 gives no guidance on a runway obstruction at any speed, or on 4.6.4 where continuing is itself hazardous.

Recurrent training syllabus TRG-SYL-07, module 4, has four scripted simulator scenarios, all aircraft system events: engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt with the take-off continued. In the 2025 to 2026 cycle, 214 crew completed module 4. No crew received an external obstruction scenario in the five training cycles on file.

### Performance

The scheduled accelerate-stop distance required was 1,780 m against the 1,847 m ASDA, a 67 m margin. It assumes the stop begins at or before V1 with symmetric reverse.

## Route risk assessment and change management

The current assessment for Tarlton Springs was route risk assessment RRA-041 issue 3 (11/2022). Its hazards included "no alternate within 200 nm at night" and "limited engineering support", and it imposed "daylight operations only pending review of night lighting and diversion fuel policy". There is no issue 4.

Change management record CMR-2025-118 introduced the evening rotation, six days per week, from 14/07/2025. This was the first scheduled company operation into Tarlton Springs after last light. The record assessed the change as "schedule change only", stated "No new hazard identified", and answered "No" to whether a route risk assessment review was required. The Head of Flying Operations approved it on 22/05/2025. It does not mention the daylight-only restriction, and RRA-041 was not reopened. The Safety Department only received the approved document.

## Related occurrences

No wildlife strike involving a company aircraft had been reported at Tarlton Springs. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area, 38 of them between one hour before and two hours after last light. Eleven recorded animals on the sealed surface of runway 09/27. The log is a paper register kept in the aerodrome office and is not reported to any operator. The reporting officer confirmed no airline had ever asked for it.

The captain said he had seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening. He had mentioned this to ground staff but not reported it to the company.

## Safety analysis

### External Influences

#### Standard's threshold left the aerodrome with no documented wildlife plan

Aerodrome Standard AS-139 section 10.4 sets two levels of obligation for wildlife hazard management at a certified aerodrome. Every operator must meet section 10.4.1: "The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." A documented plan, however, is required only above a threshold. Section 10.4.2 applies "where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year."

Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. Under section 10.4.3, its measures under 10.4.1 "need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place." The aerodrome was therefore not in breach of the standard.

Nor did it hold a wildlife hazard management plan, a hazard assessment or a dispersal programme. Its only recorded measures were the single inspection before the first arrival of the day, under Aerodrome Manual 6.3, and a sighting log that nobody reviewed. These facts were confirmed against the certified aerodrome register and the aerodrome operator's own compliance file. It is virtually certain that the aerodrome had no documented wildlife plan, and nothing indicated that one would have been held had the threshold not excused it.

Without a plan, what the aerodrome knew about its wildlife hazard stayed where it was recorded. The sighting log from 01/2025 to 06/2026 held 46 entries, of which 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office and was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

The northern perimeter fence followed the same pattern. In 04/2023, works register item WR-221 recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers. It noted animals passing through and recommended replacement at an estimated \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time recorded as unfunded and held for review at the next cycle. Item WR-263, raised in 02/2026, recorded that the same section was deteriorating further. A site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

The fence was handled as an ordinary maintenance item competing for council funds, and no hazard assessment was attached to any of the three deferral decisions. A documented plan brings a hazard assessment, an owner for the hazard, and an audit against both. Under such a plan, the northern fence would have been a recorded wildlife control, and its failure would have had to be accounted for rather than deferred as a works item. On this view, a plan would have put in front of the council the information that was missing from each budget round.

There was also evidence against this link. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures, whether or not a plan was documented. A plan is not the only means of repairing a fence or of passing a sighting log to an airline. Moreover, the council's deferrals were

budget decisions, and there was no evidence that a documented plan would have attracted funding any more successfully. A plan does not, of itself, pay for a fence.

Weighing both sides, the deferrals were made without a hazard assessment before the council, and that assessment is what a plan would have supplied. On balance, it was likely that the absence of a documented wildlife plan contributed to the failed northern fence remaining unrepaired. This finding rests on less secure ground than the other contributing factors in this report, because the evidence against it is genuine and was not discounted. The contribution of the unrepaired fence to the two kangaroos reaching the sealed runway is analysed under Organisational Influences.

The absence of an aerodrome plan was not found to have influenced the operator's criterion for requesting a runway inspection after last light. That criterion, at Appendix 8A, was the operator's own rule, and no aerodrome plan could have changed it. It is analysed under Risk Controls.

## Organisational Influences

### Failed northern fence left unrepaired through three budget rounds

The aerodrome operator's works register recorded the northern boundary fence between the 700 m and 1,300 m runway markers as failed in 04/2023 (item WR-221). The entry noted four displaced panels, strained wire and animals observed passing through, and estimated replacement at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. A further entry in 02/2026 (WR-263) recorded the same section deteriorating further. When the site was inspected on 23/06/2026, it showed a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark (Figure 1). It is virtually certain that the fence was in this condition on the night of the occurrence.



*Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026*

The pad lies on the northern side of the runway, 138 m before the point where the first carcass was recovered. Both carcasses, at 1,188 m and 1,221 m, were on the same side, and the first officer saw the animals moving onto the runway from the northern side. Together, this evidence identifies the failed section as the route by which these two kangaroos reached the movement area. Had WR-221 been actioned in 2023, the section would have been intact in 06/2026 and that route closed.

The evidence did not establish, however, that a replacement fence would have kept kangaroos out. WR-221 recommended replacement without stating a specification, and kangaroos can clear a stock fence. The pad shows that these animals came through the gap, but it does not show that an intact fence would have stopped them. The aerodrome's wildlife sighting log also does not record where any of its 41 kangaroo entries came onto the aerodrome. For these reasons, the unrepaired fence was assessed as likely to have contributed, and not higher. The aerodrome's position below the threshold for a documented wildlife plan is discussed under External Influences.

### **Evening service approved without reopening the daylight-only risk assessment**

RRA-041 issue 3, dated 11/2022, is the operator's current route risk assessment for Tarlton Springs. Its assessed hazards included "single runway with no lighting redundancy" and "no alternate within 200 nm at night", and it imposed the restriction "daylight operations only pending review of night lighting and diversion fuel policy".

CMR-2025-118 introduced a scheduled evening rotation from 14/07/2025, the first company operation into Tarlton Springs after last light. The record assessed the change as "schedule change only", stated "No new hazard identified", and answered "No" to whether a route risk assessment

review was required. The Head of Flying Operations approved it on 22/05/2025. CMR-2025-118 does not refer to the daylight restriction, RRA-041 was not reopened, and no issue 4 exists. The Safety Department's records show no involvement beyond receipt of the approved record. It is virtually certain that the evening service was approved without the restriction being reviewed. Because the approver has since left the company and could not be interviewed, the reason the change was assessed in this way could not be determined.

The change process asked whether the change was new, not whether the existing assessment still held. Had the restriction been honoured, every company departure from Tarlton Springs would have been in daylight. SB512 was scheduled off blocks at 17:20 local and left on schedule. It began its roll at 17:30, after the end of civil twilight at 17:25. The aircraft therefore rolled in darkness because of its schedule, not because of a delay.

There is evidence against this connection. The restriction names night lighting and diversion fuel, not wildlife. A review that reopened RRA-041 might have dealt with those two items and approved the evening service with lighting and fuel conditions, leaving the departure after last light unchanged. What such a review would have concluded cannot be known. Even so, lifting a daylight-only restriction would still have required an assessment of a night operation at an aerodrome with a single runway, no lighting redundancy and no alternate within 200 nm at night. Approval with the departure time unchanged was the less likely of the two outcomes. The approval was therefore likely to have contributed, through the darkness of the take-off roll. The change process could not have altered the fleet-wide criterion for runway inspections after last light, which is discussed under Risk Controls.

### **Evening rotation rostered to close five-sector duties within 45 minutes of the limit**

The rostering system shows that the evening rotation operated on 289 days between 14/07/2025 and 18/06/2026. On every one of those days, it formed the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00. Those duties were planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a maximum of 13 hours 30 minutes for five sectors from those sign-on times. On six days, a member of the rostered crew reported unfit, and each time the rotation went to a crew already on duty, extending that crew's day from three sectors to five. The occurrence day was the sixth. The crew signed on at 06:30, took off 11 hours into the duty, and had a planned duty of 13 hours. It is virtually certain that this roster pattern existed.

On the night, the roster placed a crew who described themselves as tired into a night departure 11 hours into their duty. The recorded data does not show that this affected the occurrence. The power levers reached flight idle 0.6 seconds after the reject decision, and maximum braking was established 2 seconds after it. That response is at the fast end of the operator's own fleet distribution, and there was no error of omission during the roll. The one decision a long duty could plausibly have affected was not requesting a runway inspection. Since no document prompted either crew member to consider that step, the duty could not have affected it. The roster was therefore unlikely to have influenced the outcome. Whether the crew were fatigued could not be determined, and that question is discussed separately as an unresolved matter.

The roster pattern nonetheless remains a standing exposure. Every operation of the rotation has closed a five-sector duty planned to within 45 minutes of the maximum. Each of those duties ended at an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support. The exposure is greatest when a crew member reports unfit and the sector passes to a crew whose day has just been extended. This pattern was likely to have increased risk to the evening operation.

## **Risk Controls**

## **Inspection after last light required only where a wildlife strike was reported in 24 months**

The operator's manual already contained a defence against wildlife on the runway at night. At an aerodrome listed in Appendix 8A, Part A 8.14.2 requires the pilot in command to "request a runway inspection from the aerodrome operator prior to the first departure after last light", and the pilot in command must not commence the take-off roll until the result has been passed. At other aerodromes, 8.14.3 states that "no specific action is required." It is virtually certain that the requirement did not apply at Tarlton Springs on the night, because the aerodrome was not listed and never had been.

The reason lay in the listing criterion. An aerodrome is listed only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months". No provision in Part A asks a crew to report a sighting, and no provision admits an aerodrome to the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began. As a result, the control could be applied at an aerodrome only after the kind of event it was intended to prevent had already happened there.

Yet the hazard was known at the aerodrome. Its wildlife sighting log for 01/2025 to 06/2026 held 46 entries, and 41 of them recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office and was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it. None of the entries was a strike, so none could bring Tarlton Springs onto the list.

Had the requirement applied, the crew would have requested an inspection and would have needed its result before the roll. Aerodrome staff disperse animals when they find them. It is very likely that the listing criterion contributed to the absence of a runway inspection before the departure, which is discussed under Individual Actions. One qualification remains: an inspection completed some minutes before departure would not guarantee a clear runway at the moment of the roll.

## **No training or guidance for a runway obstruction during the take-off roll**

Part B 4.6 sets the frame for a decision during the take-off roll. Between 80 kt and V1, 4.6.3 permits a reject only for listed conditions: fire or fire warning, engine failure, a configuration warning, loss of directional control, or the pilot in command's judgement that the aircraft is unsafe or unable to fly. Section 4.6.4 states: "At or above V1 the take-off shall be continued." The section gives no guidance on an obstruction seen on the runway at any speed, and it contains no amplification of 4.6.4 for a case in which continuing is itself hazardous.

Training matched the manual. Module 4 of TRG-SYL-07 has four scripted scenarios: an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt, continued. All four are aircraft system events, and none presents an obstruction, an animal or any other external hazard on the runway. In the five training cycles on file, no crew had been given an external obstruction scenario, and 214 crew completed the module in the 2025 to 2026 cycle alone. Because records before those five cycles were not available, it is extremely likely, rather than certain, that this gap existed. The captain and the first officer each said independently that every stop or go scenario they had been given involved an engine. Both also said that nothing had ever been placed on the runway in front of them in training.

The effect of this gap can be read two ways. The first officer called the animals at 07:30:31, at 116 kt, and the captain called the reject at 07:30:32, at 118 kt. With no scenario and no guidance, the

captain had about a second to decide a question he had not been prepared for. Against that, the captain stated that he knew the rule at V1 and departed from it deliberately, because he judged continuing to be worse. On that reading, obstruction training would not have changed what he knew, and the gap would concern the fleet rather than this reject.

On balance, it is likely that the absence of training and guidance contributed to the take-off being rejected 6 kt above V1. An unrehearsed decision made in about a second differs from one made against a briefed and practised frame. This finding concerns the frame available to the captain when he decided. The reject itself is analysed under Individual Actions.

### **Floor escape path lighting checked only every 24 months, not pre-flight**

On this type, the pre-flight emergency lighting test illuminates the ceiling emergency lights and the exit signs, but it neither illuminates nor monitors the floor proximity escape path segments. Those segments are checked only by a dedicated functional test at the maintenance programme's 24-month interval. It is virtually certain that this gap existed, and on VH-SBQ it concealed a defect. During the disembarkation, the first officer noticed sections of the aisle floor strip unlit while walking back through the cabin. A functional check on 24/06/2026 then found two of the six segments on the left side aft of row 16 dark on battery power with the system armed, and the fault was traced to a failed ballast common to both segments. The last functional test had been 11 months before the occurrence, within the interval. No check available to the crew or a line engineer could have detected the defect, and it could have remained undetected for up to 13 more months.

The defect played no part in the occurrence. No evacuation was commanded, and the main cabin lights were on throughout. Passengers left by the forward airstair 4 minutes 8 seconds after the aircraft stopped, without incident. It is unlikely that the monitoring gap or the defect affected the overrun or its outcome.

The gap still matters. In an evacuation with a fire indication, smoke, or a night exit over the wing, two dark segments in the aft cabin could decide whether passengers leave in an orderly way or find their route blocked. Between functional tests, the state of this lighting on any aircraft of the type is unknown. It is very likely that this monitoring gap increased risk. It was not established whether the same ballast is fitted across the operator's fleet, or how many aircraft are currently within the interval.

## **Local Conditions**

### **Departure in the logged kangaroo peak, 1 h before to 2 h after last light**

The aerodrome's wildlife sighting log for 01/01/2025 to 18/06/2026 holds 46 entries, of which 41 record kangaroos on or adjacent to the movement area. Thirty-eight of those 41 fall between one hour before and two hours after last light, and eleven record animals on the sealed surface of runway 09/27. On 18/06/2026 that window ran from 16:25 to 19:25 local, and the roll began at 17:30. The captain's account matches the log. He had seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening and never in daytime.

It was extremely likely that SB512 departed within the period of peak kangaroo activity. The window and the time of the roll are documentary, and the only reservation is that the log is a paper register whose completeness cannot be verified.

It was likely that departing in this window contributed to two kangaroos crossing onto the sealed runway. The encounter matched the pattern the aerodrome's own log describes, and a departure well outside the window would have been much less likely to meet animals on the runway. The log does have one limitation. It records sightings rather than presence, and aerodrome staff go off duty at 18:30, part way through the window, so the evening concentration may partly reflect when

somebody was there to see. On the day, the scheduled departure fell inside the window regardless of how the timetable ran.

### **Roll in darkness, 5 minutes after last light, with no moon**

The roll began at 17:30 local, on schedule, 32 minutes after sunset and 5 minutes after the end of civil twilight. Moonrise was not until 22:41, and the moon was 14% illuminated. There was no natural or town light on the strip. The only lighting was the runway edge lighting, which the crew had switched on through the pilot-activated lighting at 07:21:50. It is virtually certain that the roll took place in full darkness.

The investigator visited the site at the same time of day on 23/06/2026 and confirmed that the unlit graded strip either side of the sealed surface cannot be seen from the centreline. In that light, the first a crew can see of an animal crossing the strip is the animal on the lit surface. The first officer's account was consistent with this. On the CVR after the stop, she said the animals "came out of the dark on the left of the strip", and that "There was nothing to see and then there were two of them." In daylight, animals crossing the graded strip from the northern fence would have been in view before they reached the seal.

Some evidence pointed the other way. The first officer did see the animals at around 200 m once they were on the seal, but how long they had been on the strip before that is not known. Animals that reached the strip only seconds before the seal would have been seen late in daylight too. In addition, the captain was heads-in on the engine page after V1, as the procedure requires, so in daylight the sighting would still have rested with the pilot flying. On balance, it was likely that the darkness contributed to the take-off being rejected 6 kt above V1, because it delayed the point at which the crew could see the animals. This is held below the other contributing factors because the animals' time on the strip is not known. The scheduling decision that placed SB512 after last light is discussed under Organisational Influences.

### **Four northern edge lights above nominal intensity on the night**

A photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3. This is the section the animals crossed. The remaining lights were within tolerance. The lighting runs at one fixed setting, found at stage 3, and crews cannot vary it. No work on the lighting or its controller was recorded between the occurrence and the check. The aerodrome's maintenance programme checks lamp serviceability but does not measure intensity. It is likely that the four lights were at the measured intensity on the night, although nobody measured them at the time.

Whether that intensity delayed the detection of the animals could not be resolved. The captain recalled the lighting as "brighter than I'd want", a lit corridor with everything outside it black, and brighter edge lights would deepen the contrast with the dark strip from which the animals came. Against this, the first officer described the lighting as normal and said she saw the animals clearly once they were on the seal. She also said that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge. The captain's non-detection is explained by his heads-in duties after V1, and darkness alone explains why nothing off the seal could be seen.

The two accounts of the lighting disagree, and the only crew member who saw the animals saw them clearly. No available evidence shows that a 131% edge light, rather than the darkness itself, affected detection, and resolving the question would need evidence of that kind. The matter is therefore about as likely as not and remains unresolved. It is not carried into the findings.

### **Tailwind component during the take-off roll and stop**

A tailwind was ruled out because the aerodrome's automatic observations and the recorded aircraft data both showed a headwind component throughout the roll and the stop.

### **Flight crew fatigued at the time of the take-off**

The investigation examined whether the crew were fatigued and could not settle the question. Several points supported fatigue. The take-off was 11 hours into the duty, on the fifth sector, after a 06:30 sign-on, and the day had been extended that morning from three sectors to five. Both crew described themselves as tired, and the captain said he would have declined another sector.

Several points weighed against it. Both crew described the tiredness as ordinary and not impairing. The captain said that "for the two seconds that mattered, I was there", and the first officer described a normal five-sector day with a proper break in Port Carrick. The duty was within the company's limits, and the recorded performance showed no sign of degradation. The power levers reached flight idle within 0.6 seconds and maximum braking was established at 2 seconds, at the fast end of the fleet's distribution. The callouts were on time, and there was no error of omission on the roll.

Being tired after 11 hours is not the same as being fatigued to a degree that degrades performance. The only evidence for fatigue is the crew's own description, and their performance argues against it. Settling the question would require a detailed fatigue assessment, including sleep history and the duty pattern before 18/06. Even if fatigue had been present, the recorded data contradict each way it could have acted on this take-off. The matter is about as likely as not and remains unresolved. The risk associated with the roster is discussed under Organisational Influences.

## **Individual Actions**

### **No runway inspection requested before departing after last light**

It is virtually certain that no runway inspection was requested or offered before SB512 departed. Neither the CVR nor the CTAF recording contains any request for an inspection, any offer of one, or any mention of wildlife before departure, and the captain stated that wildlife was not discussed with anyone during the turnaround. The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local before the first arrival, with the result 'movement area clear'. No later inspection was carried out or requested. SB512 was the only departure after last light that day.

The inspection was available. Aerodrome Manual 6.3 provides for an inspection outside the scheduled time at the request of the reporting officer or a pilot in command. The reporting officer was on duty at the terminal during the turnaround and confirmed that one would have been carried out if asked.

Neither crew member considered asking, and nothing in their documents prompted them to. Operations Manual Part A 8.14.2 requires an inspection request only at aerodromes listed in Appendix 8A. Tarlton Springs was not listed, and for all other aerodromes 8.14.3 states that "no specific action is required". The captain had expected Tarlton Springs to be on the list and was surprised to find it was not. The omission therefore had no owner: no procedure asked either crew member to consider an inspection, and the aerodrome would have done one on request. The conditions under which an aerodrome is listed are discussed under Risk Controls.

An inspection would likely have changed the outcome. It would have covered the section from 1,050 m to 1,250 m, where the fence pad and both carcasses were found, and aerodrome staff disperse animals when they find them. The finding cannot be put higher than likely, because an inspection completed some minutes before the roll does not guarantee that the runway stays clear afterwards.

### **Take-off rejected 6 kt above V1**

It is virtually certain that the take-off was rejected above V1. The FDR shows V1 at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32 at 118 kt, which was 3 seconds and 6 kt past V1. Part B 4.6.4 states: "At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached."

The decision was made in a single exchange. At 07:30:31 the first officer called "Roos - roos, roos, on the left, they're on the -". One second later the captain said "Stopping. I have control." The captain did not see the animals himself, and after the stop he said, "I never had them. I was heads-in on the engine page after V one." In interview and in his air safety report, he explained his judgement that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country that offered no usable option. He chose the risk whose end he could see, and he acknowledged that the decision was outside procedure.

The decision did not arise on its own. The animals were on the runway, the roll was in darkness, and Part B 4.6 contains no guidance on an obstruction seen on the runway at any speed. Those factors are discussed under Local Conditions and Risk Controls. The edge lighting was also examined as a possible influence on the timing of detection, and that matter was not resolved.

The late reject extremely likely contributed to the overrun. The scheduled accelerate-stop distance of 1,780 m assumes that the stop begins at or before V1, and the margin between the distance required and the distance available was 67 m. The reject began 1,152 m along runway 09, which left 695 m of sealed surface, but the stopping distance actually achieved from 118 kt was 757 m. Beginning the stop 6 kt and 3 seconds late used up the 67 m margin and more, even before the loss of No. 2 reverse is counted. A late reject does not reduce the deceleration. Instead, it reduces the runway left in which to decelerate. The separate shortfall in deceleration followed from the strike and is addressed with the occurrence events.

### **Maximum braking established within 2 seconds of the reject decision**

It is virtually certain that the stop was carried out quickly. Both power levers reached flight idle 0.6 seconds after the decision, and maximum braking was established at 07:30:34, 2 seconds after it. The operator's recorded fleet mean from decision to maximum braking is 3.2 seconds. Anti-skid was active on all four wheels throughout, and brake energy reached 84% of the certified maximum without a brake fire. No. 1 reverse was selected within 4 seconds of the decision. No. 2 reverse was not available because the No. 2 propeller had been damaged in the strike. The captain identified this at 07:30:36 and called "Max brakes, max brakes." The first officer called the speeds and warned that the runway end was approaching.

The aircraft left the seal at 21 kt. It stopped 62 m into the 90 m RESA, upright and on the extended centreline, 28 m short of the end of the prepared area. A stop begun at the fleet mean response time would have carried the aircraft an estimated further 30 to 40 m. The prompt braking very likely reduced the severity of the overrun, since it is the difference between a low-speed stop on a prepared surface and the aircraft reaching the far edge of the RESA. Having found that the reject was outside procedure, the investigation also notes that the stop which followed was executed faster than the fleet average.

## **Occurrence Events**

### **Two kangaroos crossed the strip onto the sealed runway**

It is virtually certain that two kangaroos crossed the runway strip and reached the sealed surface of runway 09 during the take-off roll, and three separate lines of evidence each establish this. Two adult kangaroo carcasses were recovered on the northern half of the runway, at 1,188 m and 1,221 m from the threshold. The first officer called them on the CVR at 07:30:31, "roos, on the left", and

after the stop said they "came out of the dark on the left of the strip, crossed the centreline". The FDR recorded the impact at 07:30:33.

The first officer saw the animals moving from the northern side to the south and estimated the range at first sight as around 200 m. They came to rest on the same side as the failed northern fence, and the site inspection found an animal pad through that fence at the 1,050 m mark, 138 m before the first carcass. How the animals entered the aerodrome is a different question from whether they were on the runway, and it is considered with the fence under Organisational Influences.

The aerodrome's sighting log shows that kangaroos on the runway were not unusual. Eleven entries recorded kangaroos on the sealed surface of runway 09/27, all around last light. The movement area had not been inspected since 07:35 local that morning.

It is extremely likely that the animals' presence on the runway contributed to the occurrence. The first officer's call is the only trigger for the reject on the CVR, and the captain's "Stopping. I have control." followed one second later. The impact one second after that was with these two animals. Both the reject, analysed under Individual Actions, and the strike therefore follow directly from the kangaroos being on the runway.

## **No. 2 propeller and right main gear struck two kangaroos**

It is virtually certain that the No. 2 propeller and the right main landing gear struck both kangaroos. The impact occurred at 07:30:33 at 114 kt, one second after the reject was initiated. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Inspection after the occurrence found the following damage:

- three No. 2 propeller blades damaged beyond repairable limits, one with its tip separated
- animal tissue in two of the blade roots
- distorted right main gear doors
- two punctures in the lower right fuselage skin from propeller debris.

The recorded data, the physical damage and the recovered tissue all agree.

It is virtually certain that the strike contributed to the overrun, and its effect went beyond physical damage. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When the captain selected reverse at 07:30:36, it engaged on No. 1 only, and the CVR records the captain's "number two's not coming". The aircraft thus lost half of its reverse thrust at the start of a stop that had no margin to absorb the loss.

## **Deceleration below the scheduled accelerate-stop performance**

It is virtually certain that the deceleration achieved was below the scheduled accelerate-stop performance. The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That schedule assumes the stop begins at or before V1 with symmetric reverse, and on this occasion neither condition was met. With maximum braking and reverse on No. 1 only, the FDR shows a mean deceleration of 2.6 m/s<sup>2</sup>, approximately 15% below the value used in the certified data for this weight and runway condition. The loss of No. 2 reverse accounts for the shortfall. The recorded data and the distance reconstruction agree with the tyre marks and the nose wheel rut.

It is virtually certain that the shortfall contributed to the overrun. Distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09 when the reject began, which left 695 m of sealed surface. At the certified deceleration from that point, the reconstruction puts the stop at roughly 660 m, within the runway remaining. At the deceleration actually achieved, the

stop took 757 m. The aircraft left the sealed surface at 21 kt. Deceleration then fell to 1.9 m/s<sup>2</sup> in the gravel, and the aircraft stopped 62 m into the RESA.

The overrun was one stop with two causes. The reject 6 kt above V1, analysed under Individual Actions, began the stop later and so shortened the runway left for it. The strike then lowered the deceleration over that shorter distance. Neither alone carried the aircraft past the end of the runway, and it took both.

### **No. 2 engine power loss before the reject decision**

A No. 2 engine power loss before the reject was examined and found very unlikely. Both engines held take-off torque within 0.1% of each other, with no caution, warning or exceedance until the power levers were retarded at 07:30:32. The CVR contains no engine-related call, and the borescope found no pre-existing damage. The No. 2 power loss began at 07:30:33, with the impact.

## Findings

From the evidence available, the following findings are made with respect to the rejected take-off and runway overrun involving Saltbush Air SB512, VH-SBQ, at Tarlton Springs Aerodrome on 18 June 2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

## Contributing factors

- It is virtually certain that Tarlton Springs fell below both thresholds of Aerodrome Standard AS-139 10.4.2 and so held no documented wildlife hazard management plan, without being in breach of the standard. It is likely that this contributed by leaving the failed northern fence to be deferred as an ordinary works item with no hazard assessment behind it, although a documented plan would not of itself have funded the repair. (Safety Issue)
- The failed section of the northern perimeter fence between the 700 m and 1,300 m markers was, with virtual certainty, left unrepaired through the 2023, 2024 and 2025 budget rounds. This likely contributed by leaving open the route by which the two kangaroos reached the runway, although it was not established whether the fence as replaced would have excluded kangaroos. (Safety Issue)
- SB512 very likely departed within the peak period of kangaroo movement in the aerodrome's sighting log, which ran from 1 hour before to 2 hours after last light. This likely contributed to the animals being on the runway, although the log records sightings rather than presence and may partly reflect the hours aerodrome staff were on duty.
- It is virtually certain that the operator required a runway inspection before the first departure after last light only at aerodromes in Appendix 8A, where a wildlife strike involving a company aircraft had been reported in the preceding 24 months. It is very likely that this contributed, because Tarlton Springs had never been listed and the requirement did not apply to SB512 despite the kangaroo sightings recorded at the aerodrome. (Safety Issue)
- It is virtually certain that no runway inspection was requested before SB512 departed after last light, and that no company document prompted either pilot to consider one. It is likely that this contributed to the kangaroos being on the runway during the roll, because an inspection would have covered the section where they crossed.
- Two kangaroos crossed the graded strip from the northern side onto the sealed surface of runway 09 during the take-off roll, which is virtually certain. This very likely contributed to both the rejected take-off and the strike.
- It is very likely that neither the operator's recurrent training syllabus nor Operations Manual Part B 4.6 contained any scenario or guidance for an obstruction on the runway during the take-off roll. This likely contributed to the decision to reject above V1, although the captain stated that he knew the procedure and departed from it deliberately because he judged continuing to be the greater risk. (Safety Issue)
- The evening service into Tarlton Springs was, with virtual certainty, approved under CMR-2025-118 as a schedule change only, without reopening route risk assessment RRA-041 and its daylight-only restriction. It is likely that this contributed by placing SB512's scheduled take-off after last light, although the approver could not be interviewed and what a review would have concluded is inferred. (Safety Issue)

- With virtual certainty, the take-off roll began in darkness at 17:30 local, 5 minutes after the end of civil twilight, with the moon not yet risen. It is likely that this contributed to the rejected take-off by preventing the crew from seeing the animals until they reached the lit runway surface, although how long they had been on the strip before then was not established.
- The captain rejected the take-off at 118 kt, 3 seconds and 6 kt after V1 and contrary to Part B 4.6.4, which is virtually certain. This very likely contributed by beginning the stop with 695 m of sealed surface remaining, when the scheduled accelerate-stop distance assumed a reject at or before V1.
- It is virtually certain that the No. 2 propeller and right main gear struck the two kangaroos at 114 kt, one second after the reject began, and it is virtually certain that this contributed by making reverse on No. 2 unavailable during the stop.
- Because reverse was available on No. 1 only, the deceleration achieved was, with virtual certainty, approximately 15% below the value in the certified accelerate-stop data. This contributed with virtual certainty, as the aircraft needed 757 m to stop from the reject point against 695 m of runway remaining.

## Other factors that increased risk

- The evening rotation was, with virtual certainty, rostered to close five-sector duties planned to within 45 minutes of the 13 hour 30 minute maximum, and on six days it was reassigned to a crew already on duty. This was not found to have contributed to the occurrence. It likely increased risk on an operation into an aerodrome with no alternate within 200 nm at night. (Safety Issue)
- It is virtually certain that the floor proximity escape path lighting on the type was checked only by a functional test every 24 months and not by the pre-flight emergency lighting test, which left two failed segments on VH-SBQ undetected. This was not found to have contributed to this occurrence. It is very likely that it increased risk. (Safety Issue)

## Other findings

- The automatic weather station observations and the recorded flight data both showed a headwind component throughout the take-off roll and stop, and a tailwind component was ruled out.
- Recorded engine parameters, the cockpit voice recording and the borescope inspection showed no loss of power on the No. 2 engine before the reject decision. The No. 2 power loss began at the moment of impact.
- Maximum braking was established 2 seconds after the reject decision, against an operator fleet mean of 3.2 seconds, which reduced the distance the aircraft travelled into the runway end safety area.

## Safety issues and actions

### Standard's threshold left the aerodrome with no documented wildlife plan

Tarlton Springs Aerodrome fell below both thresholds in Aerodrome Standard AS-139 10.4.2, so it was not required to hold a documented wildlife hazard management plan. That section applies only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. The heaviest aeroplane scheduled into Tarlton Springs was 22,900 kg, and the aerodrome handled 61,400 passenger movements in 2025. Under 10.4.3, its wildlife measures did not need to be documented at all.

The aerodrome held no plan, no hazard assessment and no dispersal programme. Its only measures were the pre-first-arrival inspection at Aerodrome Manual 6.3 and a sighting log that nobody reviewed. It is virtually certain that this situation existed. The aerodrome was not in breach of the standard.

The safety significance lies in what the absence of a plan left unowned. No hazard assessment was put before the council, so the failed northern fence was treated as an ordinary works item. It was deferred three times as unfunded. It is likely that this contributed to the occurrence, but that conclusion is less firmly held than the other contributing factors. Section 10.4.1 still required the aerodrome operator to take reasonable measures. The deferrals were budget decisions, and there was no evidence that a documented plan would have secured funding for the repair. The issue arises from the standard's threshold rather than from any failure by the aerodrome operator to comply with it.

No safety action has been recorded in response to this safety issue.

### Failed northern fence left unrepaired through three budget rounds

The section of the northern perimeter fence between the 700 m and 1,300 m runway markers was, with virtual certainty, left unrepaired from April 2023 until the occurrence. In April 2023 the aerodrome operator raised works register item WR-221, which recorded four displaced panels, strained wire and animals observed passing through. It estimated replacement at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in February 2026, recorded the same section deteriorating further. When the fence was inspected on 23 June 2026, it had a continuous 140 m failure with an animal pad through it at the 1,050 m mark (Figure 1).

The failed section was the route by which the two kangaroos reached the runway. The pad lay 138 m before the point where the first carcass was recovered, on the same side of the runway. It is likely that the deferrals contributed to the occurrence by leaving that route open. However, WR-221 did not state a specification for the replacement. It was not established whether a fence replaced to that item would have excluded kangaroos, which can clear a stock fence.

In response, the aerodrome operator installed a temporary electric wire along the failed section on 25 June 2026. It also funded WR-221 from its 2026 contingency and programmed the replacement of the 140 m northern boundary fence for September 2026. This action was in progress.

### Evening service approved without reopening the daylight-only risk assessment

The operator approved a scheduled evening service into Tarlton Springs under CMR-2025-118 as a schedule change only, without reopening route risk assessment RRA-041 and its daylight-only restriction.

RRA-041 issue 3 restricted company operations at Tarlton Springs to "daylight operations only pending review of night lighting and diversion fuel policy". CMR-2025-118 was approved on 22/05/2025. It introduced the evening rotation from 14/07/2025, which was the first company operation into the aerodrome after last light. It recorded "no new hazard identified" and answered "No" to whether a route risk assessment review was required. It made no reference to the restriction, and no later issue of RRA-041 exists. The Safety Department's involvement went no further than receiving the approved change.

The operator's current risk assessment for the route and its published schedule were therefore in direct conflict, and the change process did not detect this. The process asked whether the change was new. It did not ask whether the existing assessment still held. As a result, company aircraft were scheduled to operate after last light at an aerodrome with a single runway, no lighting redundancy and no alternate within 200 nm at night, without that operation having been assessed. On 18/06/2026, SB512 left on schedule and began its take-off roll at 17:30, after the end of civil twilight at 17:25.

No safety action has been recorded in response to this issue.

## **Inspection after last light required only where a wildlife strike was reported in 24 months**

The operator required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. An aerodrome was listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision in Part A asked a crew to report a sighting, and no other evidence could admit an aerodrome to the list. Nine aerodromes were listed. Tarlton Springs never had been, and the list was last reviewed on 3 March 2025, four months before the evening service began.

This requirement was the operator's own defence against wildlife on the runway at night, and it did not apply at Tarlton Springs on the night. The aerodrome's sighting log held 41 kangaroo entries. Of these, 38 were around last light and eleven were on the sealed surface. None of them could place the aerodrome on the list, because they were not strikes and were not reported to the company. The criterion can be met only by the outcome the control exists to prevent, so no aerodrome can be listed until a strike has happened there.

### **Safety action taken**

On 26 June 2026, the Head of Flying Operations issued temporary revision TR-2026-04, which amended Appendix 8A to list Tarlton Springs. Pending a permanent change, the Part A 8.14.2 inspection requirement now applies to every departure there after last light.

### **Proposed safety action**

The investigation recommends that the operator replace the Appendix 8A listing criterion, so that an aerodrome is listed on its wildlife exposure rather than on a reported strike. It also recommends that the operator add a route risk assessment trigger for any aerodrome first operated into after last light. The operator's response was due by 30 September 2026.

## **No training or guidance for a runway obstruction during the take-off roll**

It is extremely likely that neither the operator's recurrent training syllabus, TRG-SYL-07, nor Operations Manual Part B section 4.6 prepared flight crew for an obstruction on the runway during the take-off roll. Module 4 of the syllabus contained four scripted rejected take-off scenarios, and all of them were aircraft system events:

- an engine failure at 90 kt
- an engine fire at 100 kt
- a configuration warning at 85 kt
- an engine failure at V1 plus 5 kt with the take-off continued.

No crew had been presented with an external obstruction in the five training cycles on file. Part B 4.6.4 required the take-off to be continued at or above V1. The section gave no guidance on an obstruction seen on the runway at any speed, or on a case in which continuing was itself hazardous. Both pilots stated independently that every stop or go scenario they had been given in training had involved an engine.

The issue was not confined to this crew. The current training cycle alone covered 214 crew, and none had been given the scenario. It is likely that the gap contributed to the decision to reject above V1, because the captain had about a second to decide a question he had not been prepared for. The captain also stated that he knew the procedure and departed from it deliberately because he judged continuing to be worse. On that account, obstruction training would not have changed what he knew. The gap applied across the fleet whichever reading is taken.

No safety action has been recorded for this safety issue.

## **Evening rotation rostered to close five-sector duties within 45 minutes of the limit**

The operator rostered the evening rotation (SB511 Port Carrick to Tarlton Springs and SB512 Tarlton Springs to Port Carrick) as the fourth and fifth sectors of Port Carrick-based duties. Those duties were planned to finish within 45 minutes of the 13 hour 30 minute maximum for five sectors. When a member of the rostered crew reported unfit, the rotation passed to a crew already on duty. That crew's day was then extended from three sectors to five on the morning of operation.

This was not found to have contributed to the occurrence. The recorded flight data placed the crew's response to the reject at the fast end of the operator's fleet distribution, and no step was omitted on the roll. The roster nonetheless likely increased risk.

The condition is built into the roster rather than confined to the occurrence day, so it is present every time the rotation operates. In the eleven months to 18 June 2026, the rotation operated on 289 days. Each time, it closed a duty planned at between 12 hours 45 minutes and 13 hours 15 minutes. Each of those duties ended with a sector at an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support. The exposure is sharpest when a crew member reports unfit. That happened six times in the period, and 18 June 2026 was the sixth.

No safety action has been recorded in response to this issue.

## **Floor escape path lighting checked only every 24 months, not pre-flight**

It is virtually certain that the pre-flight emergency lighting test on this type neither illuminated nor monitored the floor proximity escape path segments. Their only check was a dedicated functional test at the maintenance programme's 24-month interval.

This gap was not found to have contributed to the occurrence. No evacuation was commanded, and the passengers disembarked by the forward airstair with the main cabin lights on. It is very likely, however, that the gap increased risk.

Between functional tests, the state of the escape path lighting on any aircraft of the type is unknown. No check available to the crew or a line engineer can show it. On VH-SBQ, two segments aft of row 16 on the left side were dark without anyone being able to tell.

The circumstances would not need to change much for this to matter. A fire indication, smoke or a night evacuation over the wing would leave dark segments in the aft cabin standing between an orderly exit and a blocked one.

The safety issue is the monitoring interval rather than the defect on VH-SBQ. Rectifying one defect would not close the gap in checking. It was not established whether the failed ballast part is fitted to other aircraft in the fleet. It was also not established how many aircraft are currently within the interval.

No safety action has been recorded for this issue.

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